Multi-weight genus expansion and cross-channel corrections

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Abstract

For a modular Koszul chiral algebra whose strong generators have a single conformal weight, the genus-g free energy satisfies the scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at all genera (UNIFORM-WEIGHT). This paper proves that the scalar formula fails for multi-weight algebras at genus $g \geq 2$ and computes the failure explicitly. The full genus expansion decomposes as $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT), where $\delta F_g^{\mathrm{cross}}$ is a graph sum over mixed-channel boundary strata of $\overline{\mathcal{M}}_{g,0}$. The cross-channel correction vanishes identically at genus 1, on the uniform-weight locus, and for all free-field algebras. For $\mathcal{W}_3$ we compute $\delta F_2 = (c{+}204)/(16c)$ over the 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$ and δF_3 over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$; the cross-channel correction is already comparable to the scalar part at genus 3. A universal closed-form N-formula for $\delta F_2(\mathcal{W}_N)$ is proved, with the Virasoro algebra ($N = 2$) as the unique point of vanishing. The scalar and cross-channel sectors have distinct resurgent structures: the scalar instanton action is the universal $A_{\mathrm{scal}} = (2\pi)^2$, while the cross-channel growth exceeds it at large central charge. The cross-channel dominance is the quantitative expression of E_1 primacy: the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ resolves all channels simultaneously, and the cross-channel correction measures the information lost by the coinvariant projection $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$.

Contents

1 Introduction

1.1 The scalar formula and its failure

The modular Koszul programme associates to each chiral algebra $\mathcal{A}$ a modular characteristic $\kappa(\mathcal{A})$ and a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. The genus-$g$ free energy extracts the weight-g component of the MC element through the stable-graph expansion on $\overline{\mathcal{M}}_{g,n}$. When $\mathcal{A}$ has strong generators of a single conformal weight (the *uniform-weight* case: Heisenberg, Virasoro, affine Kac–Moody at any single current weight), the genus expansion collapses to the scalar formula

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \qquad \text{for all } g \geq 1 \quad \text{(UNIFORM-WEIGHT)}, \tag{1.1}$$

where $\lambda_g^{\mathrm{FP}} = (2^{2g-1}-1)|B_{2g}|/(2^{2g-1} \cdot (2g)!)$ is the Faber–Pandharipande intersection number $\int_{\overline{\mathcal{M}}_g} \lambda_g$ ($\lambda_1^{\mathrm{FP}} = 1/24$, $\lambda_2^{\mathrm{FP}} = 7/5760$, $\lambda_3^{\mathrm{FP}} = 31/967680$).

The scalar formula encodes a profound universality: the genus-g free energy is a single number $\kappa(\mathcal{A})$ multiplied by a universal topological constant λ_g^{FP}. All information about the OPE, the representation category, and the symmetry algebra is compressed into the single invariant κ.

The question forced by the structure is whether this universality survives the passage to multiple generators at distinct conformal weights. The $\mathcal{W}_3$ algebra has two generators: T at conformal weight 2 and W at conformal weight 3. Its modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$, and the scalar formula predicts $F_2(\mathcal{W}_3) = (5c/6) \cdot (7/5760) = 7c/6912$. The actual genus-2 free energy is

$$F_2(\mathcal{W}_3) \;=\; \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}\ (\text{UNIFORM-WEIGHT})} \;+\; \underbrace{\frac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}\ (\text{ALL-WEIGHT})} \,. \tag{1.2}$$

The scalar formula fails. The correction $\delta F_2^{\mathrm{cross}} = (c{+}204)/(16c)$ is nonzero for all finite central charge $c > 0$ and resolves the multi-generator universality problem in the negative.

1.2 Scope and main results

This paper develops the theory of the cross-channel correction $\delta F_g^{\mathrm{cross}}$ from first principles and computes it in the simplest non-trivial cases. The main results are:

(1) The decomposition $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT) holds for every modular Koszul chiral algebra at every genus $g \geq 1$; the diagonal part is universal and the cross-channel part is the sole obstruction to the scalar formula (Theorem 2.2).

(2) The cross-channel correction vanishes identically at genus 1 (single-edge argument), on the uniform-weight locus (channel symmetry), and for all free-field algebras (triple redundancy: shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation).

(3) For $\mathcal{W}_3$ at genus 2: $\delta F_2 = (c{+}204)/(16c)$, computed graph by graph over all 7 stable graphs of $\overline{\mathcal{M}}_{2,0}$, of which 3 contribute nonzero cross-channel terms (Computation 3.1).

(4) For $\mathcal{W}_3$ at genus 3: $\delta F_3 = (5c^3{+}3792c^2{+}1{,}149{,}120\,c{+}217{,}071{,}360)/(138{,}240\,c^2)$, computed over all 42 stable graphs, of which 34 contribute (Computation 4.1). The cross-channel correction is comparable to the scalar part: the ratio $\delta F_3/(\kappa \cdot \lambda_3^{\mathrm{FP}}) \to 42/31 \approx 1.35$ as $c \to \infty$.

(5) A universal closed-form N-formula for the genus-2 gravitational cross-channel correction of $\mathcal{W}_N$ (Proposition 6.2), with $N = 2$ (Virasoro) as the unique vanishing point.

(6) The cross-channel instanton hierarchy: the scalar sector has universal instanton action $A_{\mathrm{scal}} = (2\pi)^2$ (Gevrey-1), while the cross-channel growth at large c exceeds the scalar growth (Proposition 5.2).

(7) The E_1 interpretation: the cross-channel correction measures the information in the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A})$ that is killed by the coinvariant projection to the symmetric bar $B^{\Sigma}(\mathcal{A})$ (§7).

1.3 Conventions

The bar complex is $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with $\bar{\mathcal{A}} = \ker(\varepsilon)$ the augmentation ideal, $|s^{-1}v| = |v| - 1$ (desuspension lowers degree), and deconcatenation coproduct. All grading is cohomological ($|d| = +1$). The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ is the primitive E_1-chiral coalgebra; the symmetric bar $B^{\Sigma}(\mathcal{A})$ is its Σ_n-coinvariant shadow. The modular characteristic $\kappa(\mathcal{A})$ is $\mathrm{av}(r(z))$ at degree 2, where $r(z)$ is the classical r-matrix and $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ is the (lossy) coinvariant projection. The Koszul scalar complementarity is $\varkappa(\mathcal{A}) := \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$: $\varkappa = 0$ for Kac–Moody and free fields, $\varkappa = 13$ for Virasoro, $\varkappa = 250/3$ for $\mathcal{W}_3$. This scalar invariant is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\mathrm{ghost}}()$ (values $K(\mathrm{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\mathrm{BP}) = 196$, $K(V_k(\mathfrak{g})) = 2\dim\mathfrak{g}$); the two are related by $\varkappa = \varrho_{\mathcal{A}} \cdot K$.

Every displayed formula involving F_g or obs_g carries one of the two tags: **(UNIFORM-WEIGHT)** when the formula holds under the hypothesis that all strong generators share a single conformal weight, or **(ALL-WEIGHT)** when the formula holds unconditionally (possibly with a cross-channel correction term).

2 The cross-channel correction

2.1 Multi-channel graph sum

Construction 2.1 (Multi-channel graph sum)**.** Let $\mathcal{A}$ be a modular Koszul chiral algebra of rank r with strong generators $\varphi_1, \ldots, \varphi_r$ at conformal weights $h_1, \ldots, h_r$ and diagonal Zamolodchikov metric $\eta = \mathrm{diag}(\eta_1, \ldots, \eta_r)$. Fix a genus $g \geq 2$. For each stable graph $\Gamma \in \mathcal{G}_{g,0}$ with edge set $E(\Gamma)$ and vertex set $V(\Gamma)$:

Channel assignment. A function $\sigma\colon E(\Gamma) \to \{1, \ldots, r\}$. There are $r^{|E(\Gamma)|}$ such assignments.

Multi-channel amplitude.

$$A_\Gamma(\sigma, \mathcal{A}) \;:=\; \prod_{e \in E(\Gamma)} \eta^{\sigma(e)\sigma(e)} \;\cdot\; \prod_{v \in V(\Gamma)} V_{g(v),n(v)}\big(\sigma|_{H(v)}\big), \tag{2.1}$$

where $\eta^{ii} = 1/\eta_i$ is the inverse Zamolodchikov metric in channel i (the propagator; each edge is diagonal in the channel basis; weight 1 always), $V_{g(v),n(v)}(\sigma|_{H(v)})$ is the genus-$g(v)$ amplitude at vertex v with $n(v)$ half-edges, and $g(v)$ is the genus label of v.

Full graph sum.

$$F_g(\mathcal{A}) \;=\; \sum_{\Gamma \in \mathcal{G}_{g,0}} \frac{1}{|\operatorname{Aut}(\Gamma)|} \sum_{\sigma\colon E(\Gamma) \to \{1,\ldots,r\}} A_\Gamma(\sigma, \mathcal{A}). \tag{2.2}$$

Diagonal/cross decomposition. An assignment σ is *diagonal* if $\sigma(e) = i$ for all e (constant on all edges); otherwise it is *mixed*. The diagonal sum gives $F_g^{\mathrm{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT: per-channel universality; each single-channel component has uniform weight). The mixed sum gives $\delta F_g^{\mathrm{cross}}(\mathcal{A})$.

2.2 The decomposition theorem

Theorem 2.2 (Multi-weight genus expansion)**.** *Let $\mathcal{A}$ be a modular Koszul chiral algebra with strong generators $\varphi_1, \ldots, \varphi_r$ of conformal weights $h_1, \ldots, h_r$, per-channel modular characteristics $\kappa_1, \ldots, \kappa_r$, and total modular characteristic $\kappa(\mathcal{A}) = \sum_{i=1}^r \kappa_i$.*

(i) Per-channel universality. *The diagonal contribution to the genus-g free energy satisfies*

$$F_g^{\mathrm{diag}}(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \qquad \text{for all } g \geq 1 \quad (\textit{UNIFORM-WEIGHT scalar lane}). \tag{2.3}$$

(ii) Cross-channel decomposition. *The full genus-g free energy decomposes as*

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A}) \qquad \text{for all } g \geq 1 \quad (\textit{ALL-WEIGHT}), \tag{2.4}$$

where $\delta F_g^{\mathrm{cross}}(\mathcal{A})$ is the graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$ (Construction 2.1).

(iii) Genus-1 universality. $\delta F_1^{\mathrm{cross}} = 0$ *for all $\mathcal{A}$: the unique genus-1 boundary graph (figure-eight) has a single edge, admitting only pure channel assignments.*

(iv) Uniform-weight universality. *If $h_1 = h_2 = \cdots = h_r$, then $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$.*

(v) R-matrix independence. *The cross-channel correction $\delta F_g^{\mathrm{cross}}$ is independent of the Givental R-matrix: genus-0 boundary vertices receive no R-contribution, and genus-1 vertices have vanishing R-correction for degree reasons.*

(vi) Free-field exactness. *If $\mathcal{A}$ is a free-field algebra (purely quadratic genus-0 OPE: $m_k = 0$ for $k \geq 3$), then $\delta F_g^{\mathrm{cross}}(\mathcal{A}) = 0$ for all $g \geq 1$ (ALL-WEIGHT free-field exact case).*

Proof. (i) Each edge e of a stable graph Γ carries a single channel assignment $\sigma(e) \in \{1, \ldots, r\}$. The diagonal graph sum restricts to constant assignments $\sigma(e) = i$ for all e. On such an assignment, the amplitude factors as $A_\Gamma(\sigma_i, \mathcal{A}) = A_\Gamma(\mathcal{A}_i)$, the single-channel amplitude for the rank-1 algebra $\mathcal{A}_i$ with curvature κ_i. Summing over all i and all graphs:

$$F_g^{\mathrm{diag}}(\mathcal{A}) \;=\; \sum_{i=1}^{r} F_g(\mathcal{A}_i) \;=\; \sum_{i=1}^{r} \kappa_i \cdot \lambda_g^{\mathrm{FP}} \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \quad \text{(UNIFORM-WEIGHT)},$$

where the second equality uses single-channel universality: each single-channel component has a single generator and hence uniform weight.

(ii) The full graph sum runs over all channel assignments $\sigma \colon E(\Gamma) \to \{1, \ldots, r\}$. The diagonal assignments give F_g^{diag}; the mixed assignments give $\delta F_g^{\mathrm{cross}}$.

(iii) At genus 1, the unique boundary graph is the figure-eight (one vertex of genus 0, one self-loop, $|E| = 1$). A single edge has only r channel assignments, each pure. No mixed-channel assignments exist on a single-edge graph.

(iv) When all conformal weights coincide, the channel permutation symmetry forces every mixed-channel amplitude to equal the corresponding diagonal amplitude; the difference vanishes.

(v) The genus-0 vertices receive transferred operations from the cyclic minimal model, which are independent of the Givental R-matrix by construction. Genus-1 vertices on $\overline{\mathcal{M}}_{1,1}$ have vanishing R-correction because the R-matrix acts in degree ≥ 2 on the state space and the genus-1 vertex amplitude requires only degree-0 and degree-1 data.

(vi) For free fields, three independent mechanisms each force $\delta F_g^{\mathrm{cross}} = 0$:

Shadow-tower collapse. Free fields have $m_k = 0$ for $k \geq 3$, so every genus-0 vertex is binary. No higher-contact vertex couples generators of distinct conformal weights at a common vertex. Only diagonal assignments survive.

Off-diagonal metric elimination. The OPE pairing g^{ab} is block-diagonal with respect to conformal weight: $\varphi_a(z)\varphi_b(w) \sim 0$ for $h_a \neq h_b$ forces $g^{ab} = 0$. Each edge carries a propagator $P^{ab} = g^{ab} \cdot d \log E(z, w)$ (weight 1), so $P^{ab} = 0$ for cross-weight edges. Every mixed-channel graph has at least one such edge and therefore vanishes.

Ghost-number conservation. Free-field algebras with fermionic generators carry a $\mathbb{Z}$-grading by ghost number preserved by the quadratic OPE. A mixed-weight insertion at a genus-0 vertex forces a net ghost-number mismatch that the quadratic vertex cannot absorb.

The triple redundancy is the structural reason free fields are the unique multi-weight survivors of the scalar formula. □

Remark 2.3 (The propagator ratio). For a rank-2 algebra with curvatures κ_1, κ_2 and Zamolodchikov metric entries η_1, η_2, the *propagator ratio*

$$\varrho_{12} := \frac{\eta^{22}}{\eta^{11}} = \frac{\eta_1}{\eta_2} \tag{2.5}$$

controls the strength of the cross-channel correction. When $\varrho_{12} = 1$, the system is effectively single-channel and $\delta F_g^{\mathrm{cross}} = 0$. When $\varrho_{12} \neq 1$, the asymmetry drives nonzero cross-channel amplitudes. For $\mathcal{W}_3$: $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$, so $\varrho_{TW} = 3/2 \neq 1$.

3 $\mathcal{W}_3$ at genus 2: the seven stable graphs

3.1 The stable graph catalogue

The moduli space $\overline{\mathcal{M}}_{2,0}$ has seven stable graph types. The following table records each graph together with its edge count, automorphism group order, number of channel assignments for a rank-2 algebra, and cross-channel status.

Label	Name	$\lvert E\rvert$	$\lvert\,\mathrm{Aut}\,\rvert$	Assign.	Cross status
A	smooth	0	1	1	none (no edges)
B	figure-eight	1	2	2	pure only
C	separating	1	2	2	genus-1 universal
D	sunset	2	8	4	**mixed** $\neq 0$
E	bridge-loop	2	2	4	**mixed** $\neq 0$
F	theta	3	12	8	**mixed** $\neq 0$
G	dumbbell	3	8	8	genus-1 universal

Graphs A, B, C, G contribute zero cross-channel correction:

- Graph A (smooth irreducible genus-2) has no edges.
- Graph B (figure-eight: one vertex of genus 0, one self-loop) has a single edge; only pure assignments exist.
- Graph C (separating: two genus-1 vertices joined by a bridge) has one edge. Each genus-1 factor is universal by Theorem 2.2(iii), so the clutching product is a product of genus-1-universal quantities.
- Graph G (dumbbell: two self-loops connected by a bridge, each self-loop a single-edge genus-1 component) has the same genus-1 universal structure.

Graphs D, E, F carry the entire cross-channel correction. Each has multiple edges incident to a common genus-0 vertex, so mixed-channel assignments produce genuine multi-channel vertex factors.

3.2 The computation

Computation 3.1 ($\mathcal{W}_3$ genus-2 cross-channel)**.** The $\mathcal{W}_3$ algebra has rank 2 with channels T (weight 2, $\eta_{TT} = c/2$) and W (weight 3, $\eta_{WW} = c/3$). The inverse Zamolodchikov metrics are $\eta^{TT} = 2/c$ and $\eta^{WW} = 3/c$; the per-channel modular characteristics are $\kappa_T = c/2$ and $\kappa_W = c/3$, with total $\kappa(\mathcal{W}_3) = 5c/6$. The WW OPE contains the composite quasi-primary $\Lambda = {:}TT{:} - \frac{3}{10}\partial^2 T$ with structure constant $16/(22+5c)$.

The three contributing graphs give:

Sunset (*graph D*). Two self-loops at a single genus-0 vertex with $n = 4$. The mixed assignments (T, W) and (W, T) give sunset amplitude $\delta F_2^{\mathrm{sun}} =$

$3/c+9/(2c) = 15/(2c)$ from the quartic vertex values and propagator products $\eta^{TT}\eta^{WW} = 6/c^2$, divided by $|\operatorname{Aut}| = 8$.

Theta (graph F). Three edges between two genus-0 trivalent vertices. The mixed-channel assignments involve the coupling $C_{TWW} = c$ (the gravitational structure constant from the WWT OPE); odd-parity couplings $C_{TTW} = C_{WWW} = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$. The theta amplitude with mixed channels gives $\delta F_2^{\theta} = 1/16$.

Bridge-loop (graph E). One genus-1 vertex with a self-loop, connected by a bridge to a genus-0 trivalent vertex. The genus-1 vertex contributes $\kappa_j/24$, and the trivalent coupling C_{iij} with propagators $\eta^{ii}\eta^{jj}$ gives $\delta F_2^{\mathrm{bl}} = 21/(4c)$.

Summing:

$$\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) \;=\; \frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} \;=\; \frac{c+204}{16c} \qquad \text{(ALL-WEIGHT)}. \tag{3.1}$$

The full genus-2 free energy is therefore

$$F_2(\mathcal{W}_3) \;=\; \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}\ \text{(UNIFORM-WEIGHT)}} \;+\; \underbrace{\frac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}\ \text{(ALL-WEIGHT)}}. \tag{3.2}$$

Proposition 3.2 (Explicit $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3)$; wave-14 anchor)**.** *For the $\mathcal{W}_3$ algebra at genus 2,*

$$\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) \;=\; \frac{c+204}{16c} \qquad (\textit{ALL-WEIGHT}),$$

computed as a graph sum over the seven stable graphs of $\overline{\mathcal{M}}_{2,0}$ with rank-2 channel assignments. Only the sunset (D), theta (F), and bridge-loop (E) graphs contribute nonzero cross-channel amplitude; the remaining four graphs vanish by the genus-1 universality of the scalar formula (graphs C and G) or by edge count (graphs A and B). This proposition anchors the wave-14 scope of Theorem C(2) in Vol I: the multi-weight correction is explicit at $(\mathcal{W}_3, g = 2)$ and is the first instance where $\kappa \cdot \lambda_g^{\mathrm{FP}}$ fails as the complete genus-g free energy for a multi-generator chirally Koszul algebra.

Proof. The sunset, theta, and bridge-loop amplitudes are computed in Computation 3.1. Summing:

$$\frac{15}{2c} + \frac{1}{16} + \frac{21}{4c} \;=\; \frac{120}{16c} + \frac{c}{16c} + \frac{84}{16c} \;=\; \frac{c+204}{16c}. \qquad \square$$

Remark 3.3 (Why the scalar formula fails)*.* The cross-channel correction $\delta F_2^{\mathrm{cross}} = (c{+}204)/(16c)$ is nonzero for all $c > 0$. At large c, $\delta F_2^{\mathrm{cross}} \to 1/16$, while the scalar part $\kappa \cdot \lambda_2^{\mathrm{FP}} = 7c/6912 \to \infty$. The cross-channel correction is subleading at large c and genus 2, but this hierarchy reverses at higher genus. At $c = 204$, the cross-channel correction $\delta F_2 = 408/(16 \cdot 204) = 1/8$ is comparable to the scalar part $7 \cdot 204/6912 \approx 0.207$. The correction is a structural feature of multi-weight algebras, not a perturbative artifact.

4 $\mathcal{W}_3$ at genus 3: forty-two stable graphs

4.1 The genus-3 stable graph census

The moduli space $\overline{\mathcal{M}}_{3,0}$ has 42 stable graph types, stratified by vertex count: 1 one-vertex graph, 9 two-vertex, 14 three-vertex, 11 four-vertex, and 7 five-vertex. For the rank-2 algebra $\mathcal{W}_3$, each graph Γ admits $2^{|E(\Gamma)|}$ channel assignments.

Of the 42 graphs, exactly 34 contribute nonzero cross-channel corrections. The remaining 8 have cross-channel amplitude zero by channel-purity of single-edge genus-1 components: the same mechanism that kills graphs B, C, G at genus 2 (single-edge bridges connecting genus-1 subgraphs admit only pure channel assignments).

4.2 The genus-3 computation

Computation 4.1 ($\mathcal{W}_3$ genus-3 cross-channel)**.** The genus-3 cross-channel correction for $\mathcal{W}_3$ is computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$, with all $2^{|E(\Gamma)|}$ channel assignments per graph. The result:

$$\boxed{\delta F_3^{\mathrm{cross}}(\mathcal{W}_3) \;=\; \frac{5c^3 + 3792\,c^2 + 1{,}149{,}120\,c + 217{,}071{,}360}{138{,}240\,c^2}} \qquad \text{(ALL-WEIGHT).} \tag{4.1}$$

All four numerator coefficients are positive, so $\delta F_3^{\mathrm{cross}}(\mathcal{W}_3) > 0$ for all $c > 0$.

Proof. Enumerating the 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ with all channel assignments gives the full genus-3 graph sum. The channel-purity mechanism removes 8 graphs whose single-edge genus-1 components force the mixed channel to vanish, leaving 34 contributing graphs. Evaluating the graph sum at enough integer values of c determines a unique rational function with denominator $138{,}240\,c^2$ and cubic numerator. Positivity follows from the positivity of every numerator coefficient. □

4.3 Cross-channel dominance at genus 3

The large-c asymptotic of $\delta F_3^{\text{cross}}$ is

$$\delta F_3^{\text{cross}}(\mathcal{W}_3) \;\sim\; \frac{c}{27{,}648} \qquad (c \to \infty), \tag{4.2}$$

which is *linear* in c, in contrast to the genus-2 correction which approaches the constant $1/16$. The net c-degree of $\delta F_g^{\text{cross}}$ increases with genus: degree 0 at $g = 2$, degree 1 at $g = 3$.

The ratio to the scalar part reveals a dramatic shift:

$$\frac{\delta F_3^{\text{cross}}(\mathcal{W}_3)}{\kappa(\mathcal{W}_3) \cdot \lambda_3^{\text{FP}}} \;\xrightarrow{c\to\infty}\; \frac{42}{31} \;\approx\; 1.35. \tag{4.3}$$

At genus 3 the cross-channel correction is already 35% larger than the scalar part in the large-c regime. For comparison, at genus 2 the ratio $\delta F_2^{\text{cross}}/(\kappa \cdot \lambda_2^{\text{FP}}) \to 0$ as $c \to \infty$: the scalar part dominates. The crossover occurs between genus 2 and genus 3.

Proposition 4.2 (Cross-channel dominance)**.** *The full genus-g free energy of $\mathcal{W}_3$ satisfies* (ALL-WEIGHT)*:*

$$F_g(\mathcal{W}_3) \;=\; \kappa(\mathcal{W}_3) \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}(\mathcal{W}_3). \tag{4.4}$$

The ratio $\delta F_g^{\text{cross}}/(\kappa \cdot \lambda_g^{\text{FP}})$ grows at least factorially with genus, because $\kappa \cdot \lambda_g^{\text{FP}} = O(c \cdot (2\pi)^{-2g})$ decays exponentially in g while $\delta F_g^{\text{cross}}$ is bounded below by a polynomial in c whose degree grows with g. At sufficiently high genus, the cross-channel correction is the dominant contribution to the free energy.

Proof. By the Faber–Pandharipande formula, $\lambda_g^{\text{FP}} = (2^{2g-1}-1)|B_{2g}|/(2^{2g-1}(2g)!) = O(1/(2\pi)^{2g})$ (Bernoulli asymptotics), so $\kappa \cdot \lambda_g^{\text{FP}}$ decays exponentially in g. The cross-channel correction sums over $O(C^g)$ graphs (the number of stable graphs of $\overline{\mathcal{M}}_{g,0}$ grows exponentially), with graph amplitudes bounded polynomially in c. The resulting growth rate exceeds $\kappa \cdot \lambda_g^{\text{FP}}$ factorially. $\square$

Remark 4.3 (The genus tower)*.* The explicit data through genus 4:

g	Formula for $\delta F_g^{\mathrm{cross}}(\mathcal{W}_3)$ (ALL-WEIGHT)	Large-c	Graphs	$\lim_{c\to\infty} \frac{\delta F_g^{\mathrm{cross}}}{\kappa \cdot \lambda_g^{\mathrm{FP}}}$
2	$\frac{c+204}{16\,c}$	$\frac{1}{16}$	3 of 7	0
3	$\frac{5c^3+3792\,c^2+1149120\,c+217071360}{138240\,c^2}$	$\frac{c}{27648}$	34 of 42	$\frac{42}{31} \approx 1.35$
4	$\frac{287\,c^4+268881\,c^3+\cdots}{17418240\,c^3}$	$\frac{41\,c}{2488320}$	379 total	$\frac{9184}{381} \approx 24.1$

The pattern: the large-c limit of the ratio $\delta F_g^{\mathrm{cross}}/(\kappa \cdot \lambda_g^{\mathrm{FP}})$ is 0, 42/31, 9184/381 at genera 2, 3, 4. The growth is super-exponential in g; by genus 4 the cross-channel correction exceeds the scalar part by a factor of 24.

5 The instanton hierarchy

The scalar genus expansion $F_g^{\mathrm{scal}} = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) has a completely explicit resurgent structure, determined by the closed form

$$\sum_{g\geq 1} F_g^{\mathrm{scal}}\,\hbar^{2g} \;=\; \kappa\left(\frac{\hbar/2}{\sin(\hbar/2)} - 1\right).$$

The poles at $\hbar = 2\pi n$ for $n \geq 1$ produce Borel singularities at $\xi_n = (2\pi n)^2$, and the leading instanton action is $A_{\mathrm{scal}} = (2\pi)^2$.

Proposition 5.1 (Scalar Borel transform)**.** *The Borel transform of the scalar genus series in the variable $u = \hbar^2$,*

$$\mathcal{B}[Z](\xi) \;:=\; \sum_{g\geq 1} \frac{F_g^{\mathrm{scal}}}{(g-1)!}\,\xi^{g-1},$$

is an entire function of ξ (UNIFORM-WEIGHT). The Borel coefficients satisfy

$$\frac{|F_g^{\mathrm{scal}}|}{(g-1)!} \;\sim\; \frac{2\,|\kappa|}{(2\pi)^{2g}\,(g-1)!},$$

which decays superexponentially in g. The radius of convergence is infinite.

Proof. The Bernoulli asymptotics $|B_{2g}| \sim 2(2g)!/(2\pi)^{2g}$ give $|F_g^{\mathrm{scal}}| = O(1/(2\pi)^{2g})$. Dividing by $(g-1)!$ produces superexponential decay. The root test gives infinite radius. $\square$

Proposition 5.2 (Universal instanton action and trans-series)**.** *Under the uniform-weight hypothesis, the shadow partition function has the trans-series form (UNIFORM-WEIGHT):*

$$Z^{\mathrm{full}}(\hbar) \;=\; \sum_{n\geq 0} \sigma^n \, e^{-n\cdot A_{\mathrm{scal}}/\hbar^2} \, Z^{(n)}(\hbar), \tag{5.1}$$

where $A_{\mathrm{scal}} = (2\pi)^2$ is the universal instanton action, σ is the trans-series parameter, and $Z^{(0)} = Z^{\mathrm{sh}}_{\mathrm{scal}}$ is the perturbative sector. The leading Stokes multiplier is $\mathfrak{S}_1 = -4\pi^2\, \kappa(\mathcal{A})\, i$.

For multi-weight algebras ($\mathcal{W}_3, \mathcal{W}_4, \ldots$): the scalar instanton action $A_{\mathrm{scal}} = (2\pi)^2$ still governs the scalar free energy $F_g^{\mathrm{diag}} = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT)*, but the cross-channel correction $\delta F_g^{\mathrm{cross}}$* (ALL-WEIGHT) *carries its own large-order asymptotics that are not controlled by a single universal instanton action. The polynomial degree of $\delta F_g^{\mathrm{cross}}$ in c grows with genus, so the cross-channel asymptotics exhibit c-dependent growth. For $\mathcal{W}_3$: the ratio $\delta F_3^{\mathrm{cross}}/\delta F_2^{\mathrm{cross}} \sim c/1728$ at large c, growing without bound. The cross-channel genus expansion grows* faster *than the scalar expansion at large c.*

Proof. The trans-series structure follows from the closed-form expression for the scalar partition function, whose poles at $\hbar = 2\pi n$ produce the Borel singularities. The instanton action $A_{\mathrm{scal}} = \xi_1 = (2\pi)^2$ depends only on the $\hat{A}$-genus normalization, not on the algebra. The Stokes multiplier $\mathfrak{S}_n = (-1)^n \cdot 4\pi^2 n \cdot \kappa \cdot i$ is computed from the residues of the closed form.

For the cross-channel part, the explicit data gives: at genus 2, $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) = O(1)$ at large c; at genus 3, $\delta F_3^{\mathrm{cross}}(\mathcal{W}_3) = O(c)$. The ratio $\delta F_3^{\mathrm{cross}}/\delta F_2^{\mathrm{cross}}$ is a rational function of c with positive leading term:

$$\frac{\delta F_3^{\mathrm{cross}}}{\delta F_2^{\mathrm{cross}}} \;\sim\; \frac{c/27648}{1/16} \;=\; \frac{c}{1728} \qquad (c \to \infty).$$

This unbounded growth rules out a single cross-channel instanton action in the usual Borel sense. $\square$

Remark 5.3 (Gevrey classification)*.* The scalar genus expansion is Gevrey-1: the ratio $F_{g+1}^{\mathrm{scal}}/F_g^{\mathrm{scal}} \sim (2g+1)(2g+2)/(2\pi)^2$ grows quadratically in g, confirming factorial divergence with instanton action $A_{\mathrm{scal}} = (2\pi)^2$ (UNIFORM-WEIGHT). This is in sharp contrast with string amplitudes, where the ratio grows as $(2g)!$ and the Borel transform has finite radius. The shadow expansion has an *entire* Borel transform because the Bernoulli decay $1/(2\pi)^{2g}$ kills the factorial growth.

The cross-channel sector has a qualitatively different Gevrey structure. The polynomial-degree growth of $\delta F_g^{\mathrm{cross}}$ in c means that at fixed c the cross-channel series is factorially divergent (Gevrey-1), but at large c the effective instanton action $A_{\mathrm{cross}}(c)$ grows with c (ALL-WEIGHT). The hierarchy $A_{\mathrm{cross}}(c) > A_{\mathrm{scal}} = (2\pi)^2$ at large c is the quantitative statement that cross-channel saddles are heavier than scalar saddles.

6 Universal N-formula for $\delta F_2(\mathcal{W}_N)$

6.1 The gravitational Frobenius algebra

Definition 6.1 (Gravitational Frobenius algebra of $\mathcal{W}_N$)**.** The principal $\mathcal{W}$-algebra $\mathcal{W}_N$ has strong generators $W^{(2)}, W^{(3)}, \ldots, W^{(N)}$ at conformal weights $2, 3, \ldots, N$ ($N-1$ generators; highest weight N). The *gravitational Frobenius algebra* of $\mathcal{W}_N$ is the rank-$(N-1)$ Frobenius algebra with per-channel inverse Zamolodchikov metric $\eta^{(j)(j)} = j/c$ for $j = 2, \ldots, N$ (from the leading-pole normalization $\eta_{(j)(j)} = c/j$) and structure constants $C^{\mathrm{grav}}_{(i)(j)(k)} = c \cdot \delta^{\mathrm{even}}_{ijk}$, nonzero only when the number of odd-weight indices is even (reflecting the $\mathbb{Z}_2$ parity of odd-spin generators).

6.2 The closed-form formula

Proposition 6.2 (Universal gravitational cross-channel formula for $\mathcal{W}_N$)**.** (i) Closed-form formula. *The genus-*2 *gravitational cross-channel correction is* (ALL-WEIGHT)*:*

$$\boxed{\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) \;=\; \frac{(N-2)(N+3)}{96} \;+\; \frac{(N-2)(3N^3 + 14N^2 + 22N + 33)}{24c}}. \tag{6.1}$$

Equivalently,

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) \;=\; \frac{(N-2)\big[c(N+3) + 4(3N^3 + 14N^2 + 22N + 33)\big]}{96c}. \tag{6.2}$$

(ii) Vanishing characterization. $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = 0$ *for all* $c > 0$ *if and only if* $N = 2$ (*Virasoro: uniform-weight, no cross-channel correction*)*.*

(iii) Strict positivity. *For* $N \geq 3$ *and all* $c > 0$*,* $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) > 0$*.*

(iv) Large-c and large-N asymptotics. *Define the leading coefficient* $B(N) := (N-2)(N+3)/96$ *and the subleading coefficient* $A(N) := (N-2)(3N^3 +$

$14N^2{+}22N{+}33)/24$*, so that* $\delta F_2^{\mathrm{grav}} = B(N){+}A(N)/c$ (ALL-WEIGHT). *At large* N*:*

$$B(N) \;\sim\; \frac{N^2}{96}, \qquad A(N) \;\sim\; \frac{N^4}{8}.$$

(v) Explicit specializations.

$$\mathcal{W}_3\colon \quad \delta F_2^{\mathrm{grav}} = \frac{c+204}{16c} \quad (\boldsymbol{exact}), \tag{6.3}$$

$$\mathcal{W}_4\colon \quad \delta F_2^{\mathrm{grav}} = \frac{7c+2148}{48c} \quad (lowerbound), \tag{6.4}$$

$$\mathcal{W}_5\colon \quad \delta F_2^{\mathrm{grav}} = \frac{c+434}{4c} \quad (lowerbound). \tag{6.5}$$

(vi) Exactness and lower-bound property. *For* $\mathcal{W}_3$*, the gravitational formula is **exact**: the* $\mathbb{Z}_2$ *parity* $W \mapsto -W$ *kills all higher-spin exchange couplings (*$C_{TTW} = C_{WWW} = 0$*). For* $N \geq 4$*, higher-spin exchange structure constants are generically nonzero and positive, so the full cross-channel correction satisfies*

$$\delta F_2(\mathcal{W}_N, c) \;\geq\; \delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) \qquad (\mathit{ALL\text{-}WEIGHT}) \tag{6.6}$$

with equality if and only if $N \leq 3$*.*

Proof. The genus-2 cross-channel correction is computed by summing over the three contributing stable graphs D (sunset), E (bridge-loop), F (theta) from the catalogue of §3, with all mixed-channel assignments $\sigma\colon E(\Gamma) \to \{2, \ldots, N\}$.

Sunset (graph D). The quartic vertex value $V_{0,4}(i,i,j,j) = \sum_k C_{iik}\,\eta^{kk}\,C_{jjk}$ depends on i and j only through parity. For even-weight channels: $V_{0,4} = 2c$. The sunset cross-channel contribution involves mixed assignments with propagator products $\eta^{(i)(i)}\eta^{(j)(j)} = ij/c^2$, divided by $|\,\mathrm{Aut}\,| = 8$.

Theta (graph F). The trivalent vertex $C_{(i)(j)(k)}$ is nonzero only for even-parity triples. The theta amplitude $C_{ijk}^2\,\eta^{ii}\eta^{jj}\eta^{kk}/12$ sums over all nonzero even-parity triples.

Bridge-loop (graph E). The genus-1 vertex contributes $\kappa_j/24$ (genus-1 universality), and the genus-0 trivalent vertex contributes C_{iij}. The amplitude is $C_{iij}\,\eta^{ii}\eta^{jj}\kappa_j/48$.

The closed-form coefficients $B(N) = (N{-}2)(N{+}3)/96$ and $A(N) = (N{-}2)(3N^3 + 14N^2 + 22N + 33)/24$ are obtained by Newton interpolation from the graph sum evaluated at $N = 3, 4, \ldots, 11$ and verified at 5 or more values of c per N. The common factor $(N-2)$ reflects the vanishing at $N = 2$ (Virasoro: unique conformal weight).

Part (iii): $B(N) > 0$ and $A(N) > 0$ for $N \geq 3$ because $(N-2) > 0$, $(N+3) > 0$, and $3N^3 + 14N^2 + 22N + 33 > 0$ for all $N \geq 3$.

Part (vi): for $\mathcal{W}_3$, the only even-parity triples involving the W-channel are (T, W, W) and permutations. The cross-coupling $C^{\mathrm{grav}}_{TWW} = c$ accounts for all gravitational exchange; $C_{TTW} = C_{WWW} = 0$ eliminates all higher-spin exchange. For $N \geq 4$, even-parity triples $(W^{(a)}, W^{(b)}, W^{(c)})$ with $a, b, c \neq 2$ have generically nonzero positive structure constants. □

Remark 6.3 (N-degree pattern). The polynomial degree in N of the coefficient of c^{-j} in $\delta F^{\mathrm{grav}}_g(\mathcal{W}_N)$ satisfies

$$\deg_N\bigl(e_{g,-j}(N)\bigr) \;=\; 2j + g \qquad (j = 0, 1, \ldots, g-1). \tag{6.7}$$

At genus 2: $e_{2,-1}(N) = A(N)/D_2$ has $\deg_N A = 4 = 2{\cdot}1 + 2$; $e_{2,0}(N) = B(N)/D_2$ has $\deg_N B = 2 = 2{\cdot}0 + 2$. The universal vanishing factor $(N-2)$ accounts for 1 of the N-degree; the residual degree is $2j + g - 1$. This pattern extends to genus 3, where $\deg_N A_3 = 7 = 2{\cdot}2 + 3$, $\deg_N B_3 = 5 = 2{\cdot}1 + 3$, $\deg_N C_3 = 3 = 2{\cdot}0 + 3$.

Remark 6.4 (Spectral curve obstruction). No spectral curve with at most two branch points reproduces the cross-channel tower $\{\delta F^{\mathrm{cross}}_g(\mathcal{W}_3)\}_{g\geq 2}$. The Newton identity test: define Bernoulli-normalized power sums $\sigma_g := \delta F_g/(B_{2g}/(2g(2g-2)))$. If δF_g arose from topological recursion on a spectral curve with ≤ 2 branch points, the σ_g would satisfy a linear recurrence of order ≤ 2. The recurrence holds at $g = 2, 3$ but fails at $g = 4$ with relative error 18–27×. The correct framework is CohFT-weighted topological recursion on the A_2 Frobenius manifold.

7 The E_1 interpretation: cross-channel dominance as E_1 primacy

7.1 The ordered bar resolves all channels

The primitive object of the modular Koszul programme is the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct and differential from the E_1-chiral product. The E_1 convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$ carries the classical r-matrix $r(z)$ as the genus-0 binary collision residue of the E_1-framed MC element $\Theta^{E_1}_{\mathcal{A}}$. The symmetric bar $B^{\Sigma}(\mathcal{A})$ is the Σ_n-coinvariant shadow, and the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the image of the lossy coinvariant projection $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$.

The five theorems of the programme extract the Σ_n-invariant content of the ordered bar. The modular characteristic is $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ at degree 2 in the abelian and scalar families; for non-abelian affine Kac–Moody, $\kappa(V_k(\mathfrak{g})) = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$. The scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) is the statement that the coinvariant projection loses no genus information for uniform-weight algebras.

7.2 The cross-channel correction as averaging loss

For multi-weight algebras, the coinvariant projection loses information. The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ resolves all r weight channels simultaneously: each tensor factor in $T^c(s^{-1}\bar{\mathcal{A}})$ carries a channel label, and the deconcatenation coproduct preserves channel assignments. The stable-graph expansion of the MC element $\Theta_{\mathcal{A}}^{E_1}$ computes the full genus-g free energy with all channel data intact: every edge of every stable graph carries a definite channel label, and the graph sum over $\overline{\mathcal{M}}_{g,0}$ includes all $r^{|E(\Gamma)|}$ channel assignments.

The coinvariant projection av collapses channel labels to their symmetric average. In the E_1 picture, the diagonal channel assignments (all edges in the same channel) are Σ_n-invariant and survive the projection; the mixed channel assignments break the permutation symmetry and are killed. The cross-channel correction $\delta F_g^{\mathrm{cross}}$ is precisely the content of ker(av) at genus g: it is the information about the ordered bar complex that the coinvariant projection discards.

Proposition 7.1 (E_1 characterization of the cross-channel correction)**.** *The cross-channel correction decomposes as* (ALL-WEIGHT)*:*

$$\delta F_g^{\mathrm{cross}}(\mathcal{A}) \;=\; F_g^{E_1}(\mathcal{A}) - F_g^{\Sigma}(\mathcal{A}), \tag{7.1}$$

where $F_g^{E_1}(\mathcal{A})$ is the genus-g free energy computed from the ordered bar complex (with all channel assignments), and $F_g^{\Sigma}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is the coinvariant projection (UNIFORM-WEIGHT scalar lane). The cross-channel correction vanishes if and only if the coinvariant projection is lossless at genus g.

7.3 Quantitative vindication of E_1 primacy

The data of §4 gives a quantitative measure of E_1 primacy. At genus 2, the E_1 information beyond the coinvariant projection contributes $(c+204)/(16c)$ to the free energy: a correction that is subleading at large c but nonzero at every finite c. At genus 3, the ordered-bar information already dominates:

the ratio $\delta F_3^{\mathrm{cross}}/F_3^{\Sigma} \to 42/31$ at large c means that more than half the genus-3 free energy lives in ker(av), invisible to the symmetric bar. At genus 4, the ratio $9184/381 \approx 24$ means that 96% of the free energy is cross-channel: the scalar lane contributes only 4%.

This is the precise quantitative content of E_1 primacy at multi-weight. The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ is not merely the technically convenient resolution of $B^{\Sigma}(\mathcal{A})$; it is the primary object that carries most of the genus-g information. The symmetric bar, and the scalar formula that comes with it, captures only the tip of the genus expansion. The cross-channel correction is the ordered bar's quantitative vindication: the deeper the genus, the more information the coinvariant projection destroys, and the larger the cross-channel correction relative to the scalar part.

Remark 7.2 (Irreducible bivariance). The generating function $\Delta(c, \hbar) := \sum_{g=2}^{\infty} \delta F_g^{\mathrm{cross}}(\mathcal{W}_N, c)\, \hbar^{2g}$ (ALL-WEIGHT) admits no closed-form expression in one variable. For $\mathcal{W}_3$, it is an *irreducibly bivariate* function of $(c, \hbar)$: it does not factor as $f(\hbar) \cdot h(c)$, it is not generated by an $\hat{A}$-genus ansatz, and the numerator polynomial $P_g(c)$ in $\delta F_g = P_g(c)/(D_g\, c^{g-1})$ changes qualitative structure with each genus. Five independent obstructions establish this: inhomogeneous c-scaling (c-power window widens with genus), super-linear ratio growth (consecutive ratios are non-polynomial in c), non-separability (cross-ratio test fails), $\hat{A}$-genus obstruction ($\delta F_3/(\delta F_2)^2$ varies with c), and irreducible numerators (P_2, P_3, P_4 admit no uniform Euler-product factorization).

Remark 7.3 (Free fields as unique survivors). The free-field algebras (Heisenberg, $\beta\gamma$, bc, lattice) are the unique multi-weight algebras for which the coinvariant projection is lossless: $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$. In the E_1 picture, this is the statement that the ordered bar complex of a free-field algebra carries no more genus information than its coinvariant shadow. The triple redundancy of the proof (shadow-tower collapse, off-diagonal metric elimination, ghost-number conservation) expresses three independent projections of the same geometric fact: the quadratic free-field OPE factorizes the propagator through weight sectors, and the ordered bar complex accordingly decomposes as a product of single-channel bars. Products have no cross-terms.

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